Candidate-Aware and Change-Guided Learning for Remote Sensing Change Detection

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***Abstract*— Change detection (CD) in remote sensing images aims at revealing Earth surface changes between co-registered bitemporal images. A common way to reveal changed areas is to directly mix bitemporal features and generate CD results through supervised learning. However, a certain change usually corresponds to a real object in either of the two images, which exhibits coarse/fine shapes in different scales. Therefore, a coarser-to-finer method called candidate-aware and change-guided network (CACG-Net) is proposed to effectively detect changes, where candidate objects are revealed and associated with interesting changes. Specifically, there are three key components. They are multistage change decoder (MCD), candidate-aware learning (CAL), and change guidance module (CGM). MCD reveals the most important changed objects in the coarse shape from the basic features extracted by the backbone (ResNet-18). To capture changes of interest, CAL is designed to select candidate objects in each temporal image, where a segmenter is utilized with variant change-losses. CGM intends to enrich the change details step-by-step by combining coarser change results and finer features so that changed objects are gradually revealed in a coarser-to-finer way. Furthermore, deep supervision is employed throughout the layers of CACG-Net in the training procedure, which mitigates the learning difficulty in both deep and shallow layers. Test results on four popular datasets indicate that the proposed method outperforms several state-of-the-art CD algorithms in terms of accuracy and efficiency.**

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I. Introduction

AS ONE of the major remote sensing technologies in Earth observation, change detection (CD) is a process of the pixel-to-pixel task, which discoveries "significant Earth surface changes" between co-registered bitemporal images [1]. Generally, the changes include vegetation changes due to seasonal alternation, environment changes due to natural disasters (e.g., waterflood, wildfire, and earthquake), man-made facility changes due to human activities (e.g., road, building, vehicle, and bridge), and some other specific objects according to various applications [2]. As a matter of fact, it has played a valuable role in many practical applications, such as disaster assessment [3], [4], land use planning [5], [6], and urban management [7], [8].

From the early days of the CD task, many traditional methods [9] have been explored to meet the needs of different applications, such as image difference [10], [11], change vector analysis (CVA) [12], [13], and support vector machine [14], [15]. However, recently with the development of remote sensing equipment, the quality of remote sensing images has increased highly in spatial-resolution, spectral-resolution, and temporal-resolution. On the one hand, high-quality remote sensing images provide more detailed surface information in complex scenes, but on the other, they increase the difficulty of feature learning to a large extent, which makes traditional methods based on simple handcrafted features perform poorly.

With the rapid development of deep learning (DL) [16] technology, it has been widely applied and promoted in the CD field. In recent years, there have been numerous DL-based CD methods [17], [18], which can be roughly categorized into supervised and unsupervised. In this article, we focus on supervised DL methods. Such methods are able to achieve higher accuracy in CD tasks by learning labeled data. Convolutional neural network (CNN), which has the ability to capture contextual information and extract strong discriminative features, is widely used in CD tasks to identify relevant changes in remote sensing images [19]. Zhan et al. [20] first employed a fully convolutional structure to design a novel deep Siamese convolutional network for extracting hierarchical features from image pairs. These features are

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more robust and abstract compared to hand-crafted features. Fully convolutional Siamese concatenation (FC-Siam-Conc) [21] and fully convolutional Siamese difference (FC-Siam-Diff) [21] abandon the patch-based architecture in favor of an encoder–decoder UNet architecture that enables high accuracy and high inference speed. In order to mitigate the effect of illumination variations and misregistration errors, the self-attention mechanism is introduced in the spatial–temporal attention neural network (STANet) [22] to explore the spatial–temporal relations among different pixels. Furthermore, a deeply supervised attention metric-based network (DSAMNet) [23] is constructed with convolutional block attention modules to tackle the challenges of illumination, noise, and scale. For high-resolution remote sensing images, the image fusion network (IFN) [24] is designed, which contains a deeply supervised difference discrimination network for capturing the complete boundaries of changed areas. In [25], a scale-invariant learning method is proposed to meet the demand for continuous cross-resolution CD in real-world applications.

Furthermore, some others notice that long-range concepts affect CD results to a large extent. In this regard, CNN may be deficient in capturing long-range dependencies due to its local receptive field limitation. In contrast, the transformer is able to show promising performance due to its self-attentive mechanism. Transformer's self-attention mechanism is able to effectively capture long-range dependencies in images, which is crucial for understanding and comparing remote sensing images from different periods. In addition, the transformer is able to process and integrate information on a global scale to provide deeper contextual understanding. Therefore, transformers have gained widespread attention in the field of CD, and this has led to the emergence of many new methods based on it. The bitemporal image transformer (BIT) [26] uses tokens and a transformer encoder to learn rich context. In [27], a CNN-transformer network with multiscale context aggregation (MSCANet) is designed to maintain both efficiency and effectiveness for CD tasks. To achieve accurate boundary CD results, an EGDE-Net [28] consisting of an edge-guided transformer and a feature differential enhancement module is designed and implemented in a supervised way.

However, existing CD methods usually mix bitemporal features in all scales for the final result, or reveal changed objects by concatenating, subtracting, or adding bitemporal features, which results in the loss of change details such as edges and boundaries. It is noticed that a certain change exhibits different characters in different scales and it always corresponds to a real object in either of the bitemporal images. Therefore, we introduce a novel coarser-to-finer method named candidate-aware and change-guided network (CACG-Net) to reveal candidate-changed objects in different scales step-by-step. Specifically, it is achieved by three key components. They are multistage change decoder (MCD), candidate-aware learning (CAL), and change guidance module (CGM), which are designed to reveal rough changed objects, preserve the integrity of changed objects, and enrich change details, respectively. The contributions of this article are fourfold.

1) In view of that high-level coarse features containing the most important change information and consuming less computation, we construct the MCD, first of all, to coarsely reveal the most important changed objects, which combines the information from the highest three stages.
2) Considering that a certain change always corresponds to a specific target in either of the bitemporal images, the CAL is designed to select candidate objects in each temporal image, which are associated with the change of interest by variant change-losses. This facilitates revealing the overall shape of the change target.
3) Besides, CGM intends to enrich the change details step-by-step through combining coarser change results and finer features, so that changed objects are gradually revealed in a coarser-to-finer way.
4) Furthermore, both the weighted binary cross entropy (wBCE) loss and the weighted intersection over union (wIoU) loss are employed in the training procedure to put more emphasis on the edges of changed objects so that clear boundary could be obtained in CD result.

It is organized as follows for the rest of the article. Section II reviews related works about DL-based methods. Section III describes the framework of CACG-Net and details the three key components MCD, CAL, and CGM as well as the loss function. Section IV carries out the experimental analysis aimed at evaluating the effectiveness of the proposed CACG-Net. Section V discusses four aspects: parameters, model efficiency, backbones, and loss functions. At the end, Section VI summarizes the work done in this article.

II. Related Works

In recent years, vision tasks such as image segmentation [29], image classification [30], and object detection [31] have successively attempted to apply DL techniques to their tasks and have achieved significant breakthroughs. These successful precedents provide a new idea for the research of CD. For this reason, some DL-based CD methods have been successively proposed to meet the diversity of practical applications, and in the following, we will introduce some DL-based related research.

A. DL Architecture

With the rapid development of DL technology, a variety of architectures such as fully connected network (FCN) [32], [33], [34], CNN [35], [36], [37], [38], [39], recurrent neural network (RNN) [40], [41], graph convolutional network (GCN) [42], [43], and transformer [44], [45] have been successively proposed. These diverse DL architectures have injected new vitality into the development of remote sensing CD. The distinctive feature of FCN is its densely connected structure between layers, where each neuron in a layer is connected to all neurons in the preceding layer. This structure allows the network to capture rich information, but it also results in a large number of parameters, which increases the complexity of the model. Wang and Li [34] introduced a multilayer perceptron into the CNN-based CD model, constructing a

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hybrid model aimed at reducing pseudo-change interference. CNN has been widely used in CD tasks with its powerful spatial feature extraction capability. However, it may have limitations in capturing long-range dependencies due to the characteristics of the local receptive field. Zhang and Shi [37] proposed a feature difference CNN, named FDCNN, which uses the CVA to calculate the change magnitude of each pixel as prior knowledge to guide the training of the loss function. In [38], a spatial–spectral attention network for land cover CD in remote sensing images is proposed, which employs the change magnitude image as a guide image for directing the neural network to localize the changed areas. Chen et al. [39] proposed an unsupervised single-temporal CD framework, aimed at reducing the reliance on a large number of pixel-level labeled samples, thereby enhancing the practicality and efficiency of CD. RNN has a strong ability to process time-series data and is characterized by the presence of loops in the network structure, which allows it to maintain the memory of previous inputs, with the shortcoming that it has a limited ability to process longer dependencies. Bai et al. [40] used a long–short-term memory network to model the temporal correlation between bitemporal images, aiming to obtain more discriminative features. GCN is a DL architecture for processing graph data, which utilizes the connection information between nodes to update the feature representation of each node, with the shortcomings of high computational and storage requirements. In [42], a graph representation learning framework is proposed for unsupervised multimodal CD. The core of the transformer is the self-attention mechanism, which enables it to effectively capture long-range dependencies, but this comes with high computational requirements. Both BIT [26] and MSCANet [27], mentioned in Section I, adopt a strategy of combining CNN with transformer, aiming to complement the advantages of these two architectures. Yuan et al. [45] proposed a novel transformer-based Siamese network named Pyramid-SCDFormer, aimed at precisely identifying small-scale changes and fine edge details.

B. Learning Strategy

Under the DL framework, CD methods are mainly categorized into two classes, classification learning, and metric learning, based on the differences in training strategies. The classification learning strategy is to treat the CD task as a binary classification (i.e., changed and unchanged) problem, and obtain the probability that each pixel is in the changed or unchanged category through classification learning. The commonly used loss function is the cross-entropy loss. Classification learning strategies have a wide range of applications in CD tasks, such as BIT [26] and MSCANet [27] mentioned in Section I. In contrast, the metric learning strategy reflects the degree of change by measuring the distance between pixel pairs and increasing the distance of the changed pixels while decreasing the distance of the unchanged pixels in the training procedure in order to achieve distinguishability between changed and unchanged areas. The commonly used loss function is the batch contrastive loss. STANet [22] as well as DSAMNet [23] mentioned in Section I are metric learning based methods.

C. Fusion Strategy

1) Bitemporal Feature Fusion Strategy: The core of the CD task is to extract the changes, so it is crucial to accurately obtain the difference information in the bitemporal images, which makes the bitemporal feature fusion strategy a key issue in the research. Feature concatenation, feature addition, and feature subtraction are three commonly used methods to obtain bitemporal information. However, adopting them simply faces some problems. Adopting feature concatenation and feature addition may retain a large amount of background information, resulting in information redundancy. Adopting feature subtraction may lose useful information, resulting in the loss of detailed information on the edges of the changed objects. Thus, more research focuses on increasing attention to changing information while suppressing the interference of irrelevant information. Liu et al. [46] proposed an adaptive spatial and difference learning method that attenuates background information and highlights changed features by performing difference learning on fused features. Chen et al. [47] proposed a feature constraint network that combines edge and difference information by constructing summation branches and difference branches. In addition, an attention mechanism [48], [49], [50] is proposed to help the acquisition of bitemporal information. In [48], a super-resolution-based CD network (SRCDNet) is proposed, which first adds spatial and channel attention to the extracted bitemporal features in order to capture more effective information before acquiring difference information.

The essence of some methods mentioned above is to increase the focus on the change information in the bitemporal fusion features, but the effect they bring is limited. The reason is mainly attributed to two points, one is insufficient filtering of background information, and the other is insufficient revealing of change detail information. For this reason, we construct a CAL strategy on each temporal feature before acquiring the bitemporal fusion information. The CAL process is used to select candidate objects on each temporal image and associate them with the change of interest by variant change-losses. This stems from the basic fact that a certain change always corresponds to a specific target in either of the bitemporal images. With CAL, candidate objects can be separated from a large amount of background information, maintaining the integrity of the changed objects. In addition, to focus more on the edge details, the weight of the edges is increased in the training procedure. The validity of CAL is demonstrated in Section IV.

2) Multilevel Feature Fusion Strategy: The shallow layer of the neural network is able to learn low-level features with precise location information, while the deeper layer is able to learn high-level features with rich semantic information. It is obvious that combining different levels of features can acquire richer features, which is certainly helpful in accurately recognizing changes. Therefore, multilevel feature fusion strategy becomes another important issue for research in the field of CD. Fang et al. [51] proposed a densely connected network that facilitates information transfer across different levels via dense skip connections. In addition, an ensemble channel attention module is constructed to fuse the features of different

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levels. In [52], a difference-enhancement dense-attention CNN (DDCNN) is proposed, which achieves the aggregation of high-level features to low-level features by means of high-level feature guidance. In [53], a multilevel incremental aggregation strategy is proposed to achieve fusion between different levels of features and to supervise the resulting intermediate features.

Unlike several of the above methods that focus only on fusing features at different levels, the method proposed in this article hopes to utilize the coarser change results produced at the higher levels to guide the learning of finer features produced at the lower levels, thus gradually revealing the changed objects in a coarser-to-finer way. CGM is proposed for realizing the above process, which effectively preserves the changed objects in coarse scale and reveals the change details in fine scale. The validity of the CGM is also demonstrated in Section IV.

III. Methodology

A. Framework

The framework of the proposed CACG-Net is illustrated in Fig. 1, where ResNet-18 with five stages (Stage1, Stage2, ..., Stage5) is utilized as backbone [54]. Stage1 consists of a 7×7 convolution layer and a max-pooling operation, and the remaining four stages are residual blocks. The output size of Stage2 is the same as that of Stage1, which is half the size of the original input. While in the following stages, the output size is progressively halved. Moreover, there is a receptive field block (RFB) [55] employed in each stage, which takes account of the varying object size and generates feature maps with the same channel number. Note that the proposed operators in this article are mainly conducted on the highest four stages (i.e., Stage2–Stage5), as shown in Fig. 1. Therefore, unless otherwise specified, each stage or all stages mentioned in the following discussion refers to Stage2–Stage5. In general, the proposed CACG-Net generates coarser-to-finer CD results along with multiscale features from different stages.

As illustrated in Fig. 1, there are three key components in the proposed CACG-Net, i.e., MCD, CAL, and CGM, which are marked in yellow, pink, and green blocks, respectively, and the input of each component is indicated by the arrows in the same color. It is noticed that the most important changed objects are revealed by MCD to obtain rough CD results, where high-level features are combined together and a CD loss function L_6 is employed. In order to reveal the overall shape of the changed target, a segmenter with variant change-losses is designed in the proposed CAL, which selects candidate objects in each single-temporal image. Therefore, the interesting areas associated with changed objects could be captured as a whole. Then, several CGMs utilize the coarser CD result to generate the finer one step-by-step, where the coarser CD map is employed to learn more change details from the finer feature map.

In the training procedure, there are several supervised losses related to the output of MCD and CGMs (i.e., $L_6, L_5, \ldots, L_2$), which are marked with blue arrows in Fig. 1 and correspond to change maps in different scales. After training, the final CD result is obtained from the last CGM output in the finest scale (i.e., Stage2).

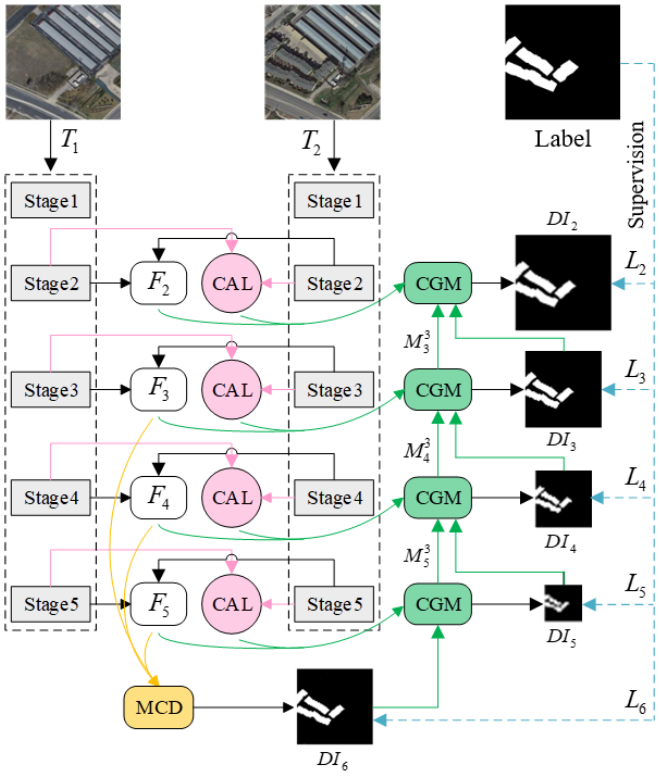


Fig. 1. CACG-Net framework.

B. Multistage Change Decoder

With the backbone ResNet-18, multiscale feature maps are generated as mentioned above. Suppose that two bitemporal images captured at different times (T_1 and T_2) are denoted by X^1 and X^2, respectively. Then, extracted features from five stages are denoted by F_l^t, where $t = 1, 2$ indicates temporal dimension and $l = 1, 2, \ldots, 5$ indicates stage index. It is noticed that coarse feature maps at high levels contain the most important change information and consume less computation. Motivated by Fan et al. [56], we combine the highest three stages ($l = 3, 4, 5$) information to generate the coarse change map by MCD, which locates the changed objects in rough shape and uses low computation.

As preparation, in each stage, the fused feature map is first obtained by difference enhancement and channel attention, as illustrated in Fig. 2. In detail, difference information W_l^d is calculated by a subtraction layer and a 1×1 convolution layer with Sigmoid activation. Meanwhile, bitemporal feature maps are concatenated in channel dimension and then two convolution layers with BatchNorm and ReLU activation are employed to generate the original fused feature map F_l^c. Then, a global pooling layer follows with a multilayer perceptron, which produces the channel attention coefficient W_l^c. The fused feature map is then updated by multiplying W_l^d and W_l^c and represented by F_l, as shown in Fig. 2.

To capture consistent semantic across layers, multistage change information is aggregated by addressing neighboring

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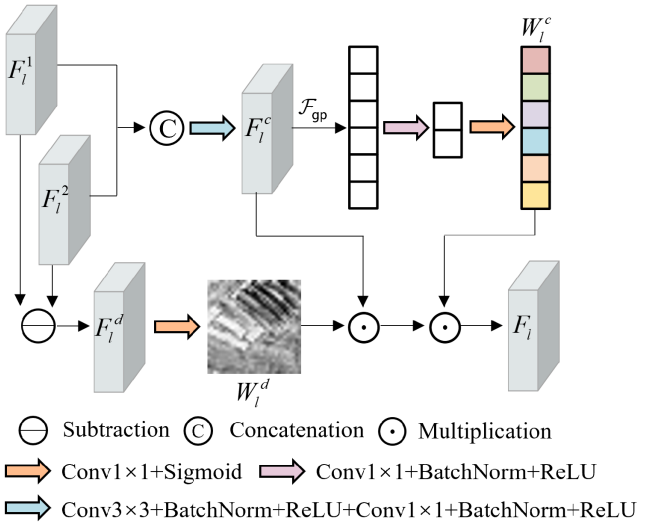


Fig. 2. Fused feature map. $\mathcal{F}_{gp}(\cdot)$ represents the global average pooling operation.

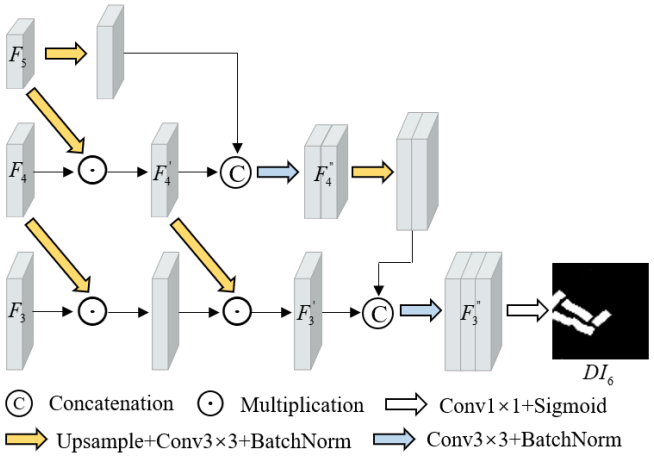


Fig. 3. Structure of MCD.

layers. The structure of MCD is illustrated in Fig. 3, which reveals the most important changed objects by decoding the F_l in the highest three stages (i.e., $l = 3, 4, 5$).

In Fig. 3, each fused feature map F_l in higher-level is upsampled (two times) and combined with the neighboring lower-level one by convolution to obtain refined one F_l', as formulated in (1). Here conv$(\cdot)$ indicates a 3×3 convolutional layer with a batch normalization, $\delta_{\uparrow}^2(\cdot)$ represents the upsampling operation and $\odot$ represents element-wise multiplication

$$\begin{cases} F_5' = F_5 \\ F_4' = \text{conv}\left(\delta_{\uparrow}^2(F_5)\right) \odot F_4 \\ F_3' = \text{conv}\left(\delta_{\uparrow}^2\left(F_4'\right)\right) \odot \text{conv}\left(\delta_{\uparrow}^2(F_4)\right) \odot F_3. \end{cases} \tag{1}$$

Specifically, each F_l' in coarser scale is upsampled followed by a convolutional layer and then concatenated with the one in finer scale following by another convolutional layer, where cat$(\cdot)$ indicates a concatenation operation, as listed in (2). The function sigm$(\cdot)$ indicates a sigmoid activation with a linear projection, which reduces the channel dimension to 1 and converts the coarser feature map F_3'' into the interval $[0, 1]$, where 0 and 1 correspond to unchanged and changed pixels, respectively. After training, the change map DI_6 is output, which contains the most important changes in the coarse scale

$$\begin{cases} F_5'' = F_5' \\ F_l'' = \text{conv}\left(\text{cat}\left(F_l', \text{conv}\left(\delta_{\uparrow}^2\left(F_{l+1}''\right)\right)\right)\right), \quad l = 3, 4 \\ DI_6 = \text{sigm}\left(F_3''\right). \end{cases} \tag{2}$$

C. Candidate-Aware Learning

Considering that a certain change always corresponds to a specific target in either of the bitemporal images, such as a building newly built in T_2 or a building built in T_1 but damaged in T_2. However, the traditional way to fuse bitemporal images by direct addition or concatenation operation may mix the target and make it hard to recognize. Therefore, we separately construct a CAL process in each temporal feature to maintain the integrity of the changed objects, which corresponds to the pink circle in Fig. 1. The principle of CAL is illustrated in Fig. 4.

As shown in Fig. 4, the main component in CAL is a segmenter Seg$(\cdot)$ with simple convolution layers, where the input is the obtained feature map F_l^t (as mentioned in the first paragraph of Section III-B). Note that the stage index l is omitted here for simplicity and the output is denoted by $S^t = \text{Seg}(F^t), t = 1, 2$.

Notice that the segmenter intends to select candidate objects in each single-temporal feature map, which is beneficial for revealing the whole shape of the change target. To achieve this, three key types of variant change-losses are designed, i.e., Sel_loss, Sub_loss, and Mul_loss, which makes the segmenter generate 1 for candidate objects and 0 for the background. Correspondingly, three simple operations "Select," "Subtract," and "Multiply" are utilized, as listed in the following equation:

$$\begin{cases} \text{Select}\left(S^1, S^2\right) = \max\left(S^1, S^2\right) \\ \text{Subtract}\left(S^1, S^2\right) = \text{abs}\left(S^1 - S^2\right) \\ \text{Multiply}\left(S^1, S^2\right) = S^1 \odot S^2. \end{cases} \tag{3}$$

There are two common observations about the change result. 1) For changed areas, there must be a candidate object in one (and only one) of the single-temporal images, i.e., there are Multiply$(S^1, S^2) = 0$, Subtract$(S^1, S^2) = 1$, and Select$(S^1, S^2) = 1$. 2) For unchanged areas, there must be both candidate objects or both backgrounds in bitemporal images, i.e., there is Subtract$(S^1, S^2) = 0$. Therefore, the variant change-losses in CAL are formulated as (4), where BCE$(\cdot, \cdot;$ weight$)$ indicates binary cross entropy loss with the weight parameter. The first two placeholders "$\cdot$" denote the prediction and the corresponding reference, respectively. During training, the area of concern (i.e., changed or unchanged) is different for the three variant change-losses, so the parameter weight is used as the index of changed or unchanged pixels

$$\begin{cases} \text{Sel_loss} = \text{BCE}\left(\text{Select}\left(S^1, S^2\right), Y; Y\right) \\ \text{Sub_loss} = \text{BCE}\left(\text{Subtract}\left(S^1, S^2\right), Y\right) \\ \text{Mul_loss} = \text{BCE}\left(\text{Multiply}\left(S^1, S^2\right), 1 - Y; Y\right). \end{cases} \tag{4}$$

In Sel_loss and Mul_loss, only the changed pixels are taken into account, so the Ground truth Y is chosen as the weight

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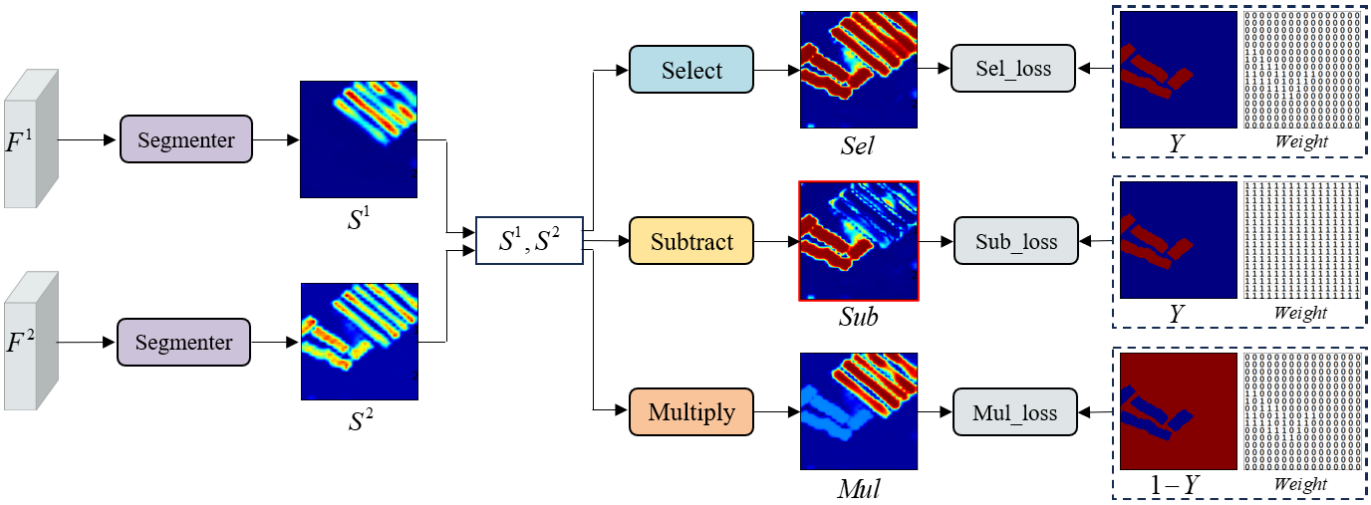


Fig. 4. Principle of CAL. The segmenter block corresponds to the function Seg(·), the Select, Subtract and Multiply blocks correspond to the functions in (3), the Sel_loss, Sub_loss, and Mul_loss correspond to the loss functions in (4) and the rightmost columns indicate the corresponding ground-truth and weight. In the following procedure, Sub marked in red square frame is utilized as the output of CAL and also as pseudo change guidance.

(1 for changed pixels and 0 for unchanged pixels). In Sub_loss, all pixels are taken into account, and the weight is an all-1 matrix at this point, so the weight parameter is omitted here. Moreover, for Mul_loss, the notation $1 - Y$ generates 0 for changed pixels to satisfy the observation above.

After training, the segmenter outputs a rough segmentation result for each single-temporal image, where some candidate objects corresponding to potential changes are revealed as a whole. The detailed learning process of CAL is clearly illustrated in Fig. 4, where the outputs of the segmenter (i.e., S^1 and S^2), the outputs of (3) (i.e., Sel, Sub, and Mul), the reference maps for each loss in (4) (i.e., Y, Y, and $1 - Y$) and the corresponding weight maps are listed, respectively, for better understanding. From the S^1 and S^2, it shows that the candidate objects are revealed in bright green or dark red (with a larger value) and the background is revealed in dark blue (with a smaller value), which are of great help in distinguishing changes in the following procedure. The outputs Sel, Sub, and Mul from three operations listed in Fig. 4 verify that change candidate objects are clearly separated from the unchanged ones, where the candidate objects show strong ability in learning changes. The rightmost two columns in Fig. 4 list the reference and the weight in different losses, in accordance with (4).

With CAL, it helps preserve more discriminative information in the learned feature map F^1 and F^2 and then separate candidate objects from the background. Therefore, the fused feature map in Fig. 2 would also contain candidate-aware information and benefit the next CGM process. In the following, the output Sub in each stage is used as pseudo-change guidance. Note that since the output Sub contains only change information, and the outputs Sel and Mul both contain unchanged objects, only Sub is used as pseudo change guidance for the CGM.

D. Change Guidance Module

As discussed before, the global change map DI_6 derived from MCD contains the most important change information and rough change location, and the output of CAL Sub_l contains pseudo change information in the lth stage. To reveal details about the changed areas, the CGM is introduced to generate finer CD results and employed in all stages as illustrated in Fig. 1. It utilizes the coarser CD result DI_{l+1} and the finer pseudo change map Sub_l to guide the learning process. The structure of CGM is demonstrated in Fig. 5.

In general, the inputs of CGM are as follows: finer feature map F_l in Fig. 2, coarser fused feature map M^3_{l+1} derived from the higher-level CGM, pseudo change map Sub_l from CAL, and predicted coarser change map DI_{l+1} from higher-level, where $l = 5, 4, 3, 2$, as shown by the red arrow in Fig. 5. As a special note, in the fifth stage ($l = 5$), there is no fusion of M^3_6 with F_5, and only DI_6 and Sub_5 guide the learning of F_5. As shown in Fig. 5, in order to reveal more details about changed objects, M^3_{l+1} is upsampled accordingly and concatenated with F_l to generate M^0. The outputs are denoted by M^3_l and DI_l in the lth stage, where the former is fed into the next CGM in lower-level, and the latter corresponds to a finer change result and is utilized to calculate the change loss.

To investigate change details purposefully, there is a change guidance operation $G(\cdot)$ repeated three times in CGM, as shown in Fig. 5. Details about the change guidance operation $G(\cdot)$ are illustrated in Fig. 6. It erases the predicted background in a coarse scale so that the following network can focus on discovering and highlighting boundary details around the changed objects. Clearly, an upsampling operation is necessary for the coarser change map DI_{l+1} to match the size of the current stage, which is denoted by N^0.

For each change guidance operation $G(\cdot)$, the feature map M^i and the predicted change map N^i are updated in the following way, where $i = 0, 1, 2, 3$. Details are illustrated in Fig. 6. First, M^i is split into several groups in channel dimension, where each group is concatenated with the previous change map N^i. These groups are concatenated together again and represented by F_{MN}, where a convolution layer with a shortcut is employed to generate the updated feature map M^{i+1}. The process is fully formulated in (5). After that, another convolution layer is used to generate a map with one

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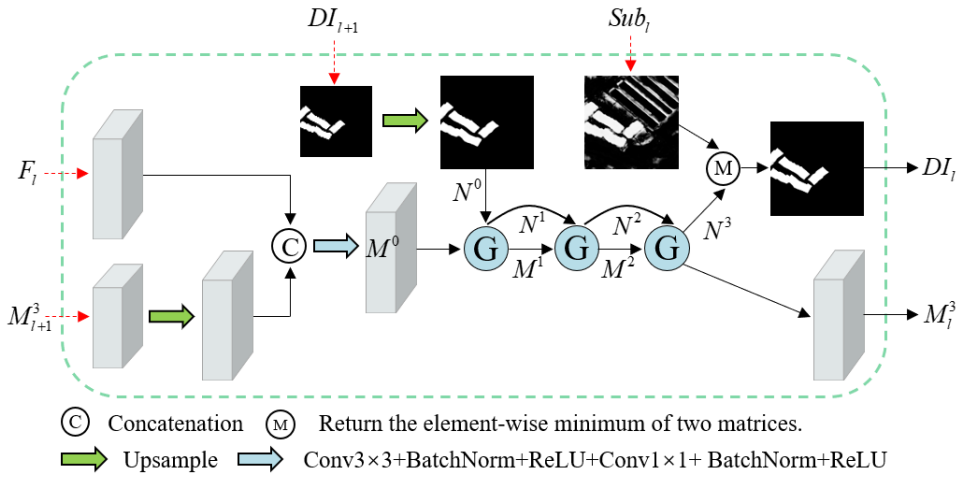


Fig. 5. Structure of CGM. Features in both finer and coarser scales (i.e., F_l and M^3_{l+1}) are concatenated to reveal more details about changed objects, where change information in both finer and coarser scales (i.e., Sub_l and DI_{l+1}) is employed to guide the learning process. The blue circles with the capital letter "G" indicate the guidance operation $G(\cdot)$.

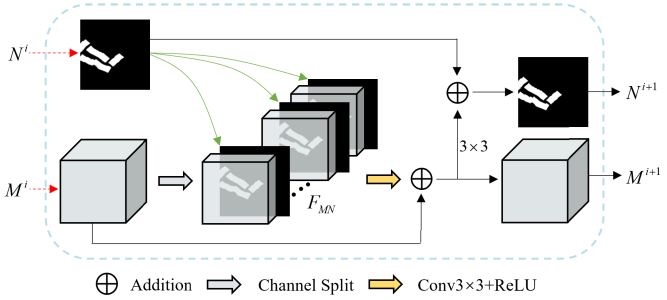


Fig. 6. Change guidance operation. The guidance operation is repeated three times in the CGM. For the first time, M^i is split into 1 group in channel dimension, for the second time it is split into 4 groups, and for the third time it is split into 32 groups, with the guidance operation going from weak to strong.

channel. It is added to the previous change map N^i, deriving the next predicted change map N^{i+1}

$$\begin{cases} M^{i+1} = M^i + \text{conv}(F_{MN}) \\ N^{i+1} = \text{sigm}\big(N^i + \text{conv}\big(M^{i+1}\big)\big). \end{cases} \tag{5}$$

Fine-grained shallow features have precise location information, but they may contain noise. Coarse-grained change maps contain the most important change information, but they have coarse change locations. Therefore, through the change guidance learning process described above, a large amount of irrelevant background information in M^i is filtered out using the coarse change information from the previous stage so that the following network can focus on discovering and highlighting the details of the boundaries around the changed objects. At the end of each change guidance operation, the change details will be enriched, thus realizing the transition of change information from coarser to finer. The above change guidance process is significantly different from the methods [37] and [38] mentioned in Section II. In [37], the change magnitude of each pixel is used as prior knowledge to train the loss function. In [38], the change magnitude image is combined with bitemporal images as input to guide the neural network in localizing changed areas, but this guidance may not be sufficient.

Following the last change guidance operation, the output N^3 is further calibrated using the finer pseudo change map Sub_l. Specifically, a Minimization operation $\text{Min}(\cdot)$ is conducted on the output N^3 and Sub_l, which returns the smaller number and generates the predicted change result DI_l in the lth stage, as illustrated in Fig. 5 and shown in the following equation:

$$DI_l = \min\big(N^3, \text{Sub}_l\big). \tag{6}$$

In the proposed CACG-Net, CGM is employed next to CAL and utilized in every stage, as demonstrated in green blocks in Fig. 1. The outputs of CGM (M^3_l and DI_l) are upsampled as input of the next CGM, and DI_l is supervised by CD Ground truth in the training procedure. After training, the output DI_2 from Stage2 CGM reveals the finest change information and generates the final CD result by upsampling.

E. Training Procedure

As discussed above, MCD derives the most important but rougher changed objects in the highest stage. CAL reveals the candidate objects in each single-temporal image, which maintains the integrity of the changed objects. Then, CGMs generate finer CD results with the help of coarser change information and finer candidate-aware features. There are mainly two types of losses associated with the training procedure, i.e., candidate-aware loss and CD loss. The former corresponds to the CAL loss listed in (4) and is summarized as follows:

$$L_l^{\text{CAL}} = \text{Sel_loss} + \text{Sub_loss} + \text{Mul_loss}. \tag{7}$$

The latter (CD loss) corresponds to the MCD and CGM modules and helps reveal the final CD result. To preserve changed objects efficiently on the coarse scale and reveal change details effectively on the fine scale, both wBCE loss and wIoU loss are employed in CD loss. That is, the CD loss is achieved by

$$L_l^{\text{CD}} = L_l^{\text{wBCE}} + L_l^{\text{wIoU}}. \tag{8}$$

In order to get clear edges in CD result, boundary pixels of changes are emphasized in the training procedure by the

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weight map constructed in (9), where the weight map of wBCE and wIoU in the lth stage is denoted by Ω_l and obtained from the boundary map E. Specifically, change boundaries or edges are simply estimated from the original CD Ground truth Y, where $\text{avg}(\cdot, k)$ indicates the average pooling function with the kernel size of k and λ_l is a positive balance parameter

$$\begin{cases} E = \text{abs}(Y - \text{avg}(Y, k)) \\ \Omega_l = 1 + \lambda_l * E. \end{cases} \tag{9}$$

In Fig. 7, a CD Ground truth map along with the corresponding edge map E and weight map Ω_l is listed as an illustration. It is clear that boundaries and edges of changed objects are represented by large values in E, where the corresponding pixel weights are set to be $1 + \lambda_l * E$. Notice that in the lower stage, there are many more details than in the higher stage, so λ_l should be larger. Conversely, coarse information plays a more important role in the higher stage, so λ_l in the higher stage should be smaller.

Thus, the wBCE loss (i.e., L_l^{wBCE}) and the wIoU loss (i.e., L_l^{wIoU}) could be formulated in (10) and (11), respectively, where $l = 2, 3, 4, 5, 6$. Note that in each stage, DI_l is properly upsampled by $\delta_\uparrow(\cdot)$ to match the size of Y

$$L_l^{\text{wBCE}} = \text{BCE}\big(\delta_\uparrow(DI_l), Y; \Omega_l\big) \tag{10}$$

$$L_l^{\text{wIoU}} = 1 - (I_l + 1)/(U_l - I_l + 1). \tag{11}$$

In (11), I_l represents the number of correctly predicted change pixels and U_l represents the number of all predicted change pixels, where pixels around boundaries are highlighted by the weight W_l, as formulated in the following equation:

$$\begin{cases} I_l = \text{sum}\big(\big(Y \odot \delta_\uparrow(DI_l)\big) \odot \Omega_l\big) \\ U_l = \text{sum}\big(\big(Y + \delta_\uparrow(DI_l)\big) \odot \Omega_l\big). \end{cases} \tag{12}$$

In the training procedure, the CD loss and the candidate-aware loss work together by two balance parameters α_l and β_l, as formulated in (13). At the very beginning, the CAL helps a lot in revealing potential candidate objects and plays an important role in training. However, as the epoch increases, the training procedure should pay more attention to distinguishing changes rather than revealing candidate objects. Therefore, the weight of the candidate-aware loss is reduced by the factor [1 − (epoch/t_epoch)], where epoch represents the current epoch and t_epoch represents the total epoch

$$L = \sum_{l=2}^{6}\big(\alpha_l * L_l^{\text{CD}}\big) + \left(1 - \frac{\text{epoch}}{t_\text{epoch}}\right) * \sum_{l=2}^{5}\big(\beta_l * L_l^{\text{CAL}}\big). \tag{13}$$

After training, the finest output C_2^{CGM} in Stage2 is upsampled by bilinear interpolation to the same size as the input image. Values in C_2^{CGM} higher than a threshold τ are recognized as change pixels and conversely as unchanged pixels.

IV. Experiments

A. Datasets

Four public datasets, LEVIR-CD [22], BCDD [57], SYSU-CD [23], and CLCD [27], which are commonly used in CD tasks, are used in the following to validate the effectiveness of the proposed CACG-Net. The LEVIR-CD and BCDD focus on building-related changes, while the SYSU-CD and CLCD contain information on changes in multiple categories. A brief description of the four datasets is given below:

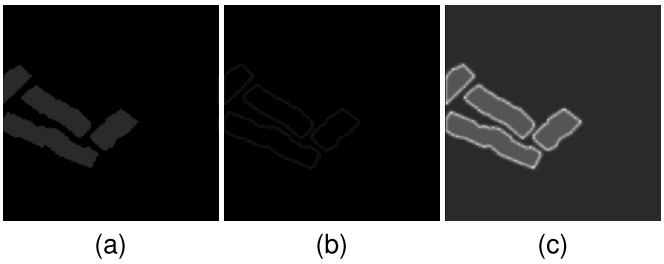


Fig. 7. Weight map illustration. (a) Y. (b) E. (c) Ω_l.

1) LEVIR-CD Dataset [22]: This dataset is a building-related dataset covering many types of buildings. It contains 637 pairs of 1024 × 1024 pixel images documenting land use in several cities in Texas, USA. The images are taken between 2002 and 2018 and have a spatial resolution of 0.5 m. The dataset division criteria provided by the authors of the LEVIR-CD dataset are followed in the experiments. Considering the memory constraints, the 1024 × 1024 pixel image is cropped into 16 nonoverlapping patches of 256 × 256 pixels. According to this cropping strategy, it makes the training set expand to 7120 pairs of images, the validation set to 1024 pairs of images and the test set to 2048 pairs of images.

2) BCDD Dataset [57]: This dataset is also a building dataset that documents building changes in the Christchurch, New Zealand area in 2012 and 2016. It contains a pair of 32 507 × 15 354 pixel images that have a spatial resolution of 0.2 m. Since the authors of the BCDD dataset do not provide dataset division criteria, we follow the dataset division scheme provided by BIT [26]. The 32 507 × 15 354 pixel image is cropped into 256 × 256 pixel nonoverlapping image patches, yielding a total of 7620 image pairs. These images are randomly divided into the training set, validation set, and test set, where the training set includes 6095 pairs of images, the validation set includes 762 pairs of images and the test set includes 763 pairs of images.

3) SYSU-CD Dataset [23]: Unlike the above two datasets, the SYSU-CD dataset contains information on changes in other categories such as vegetation, roads, and sea construction in addition to changes in buildings, which increases the difficulty of detection. The dataset is collected in Hong Kong and contains 20 000 pairs of 256 × 256 pixel images. The images have a spatial resolution of 0.5 m taken between 2007 and 2014. The 20 000 pairs of images are divided in the experiment according to 12:4:4, where the training set is 12 000 pairs of images, and the validation set and test set are 4000 pairs of images each.

4) CLCD Dataset [27]: Like the SYSU-CD dataset, the CLCD dataset contains information on changes in several categories, such as lakes, roads, buildings, and bare soil lands. The dataset is collected by the Gaofen-2 in Guangdong Province and contains 600 pairs of 512 × 512 pixel images of sample cropland changes. The images have a spatial resolution

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of 0.5–2 m and were taken in 2017 and 2019. The authors of the CLCD dataset provide the criteria for the division of the dataset, with the training set containing 320 pairs of images, and the validation and test sets containing 120 pairs of images each.

B. Comparative Methods

In the following, eight representative CD methods are introduced and used as comparison methods to verify the effectiveness of the proposed CACG-Net, i.e., FC-Siam-Conc [21], FC-Siam-Diff [21], SRCDNet [48], DSAMNet [23], BIT [26], MSCANet [27], DDCNN [52], and DMINet [53]. Here are some details.

1) FC-Siam-Conc [21]: A CNN-based Siamese network that adopts feature concatenation to obtain bitemporal information.

2) FC-Siam-Diff [21]: FC-Siam-Diff and FC-Siam-Conc have a similar structure, the only difference being the different ways of obtaining bitemporal information. FC-Siam-Diff adopts feature differences to obtain bitemporal information.

3) SRCDNet [48]: A metric learning-based network generates more discriminative features by adding local and global attention to the multilevel features extracted by the encoder.

4) DSAMNet [23]: DSAMNet also adopts a metric learning strategy to train the network. Compared to SRCDNet, it adds supervision to the hidden layer.

5) BIT [26]: A CNN and transformer-based network that utilizes CNN to extract rich bitemporal features and transformer to capture contextual information.

6) MSCANet [27]: MSCANet is also a CNN and transformer-based network that achieves more accurate detection by combining CNN and transformer. Moreover, it constructs multiple prediction heads to strengthen supervision for the deeper layers.

7) DDCNN [52]: A U-Net-based network that achieves multilevel feature aggregation by means of high-level feature guidance.

8) DMINet [53]: A multilevel intertemporal network that achieves information exchange between bitemporal features by constructing joint attention modules, and aggregation of multilevel features by utilizing an incremental aggregation strategy.

C. Evaluation Metrics

The performance of the proposed CACG-Net is evaluated using five widely used evaluation metrics in CD tasks: precision (Pre), recall (Rec), $F1$-score ($F1$), intersection over union (IoU), and Kappa. True positive (TP) and false positive (FP) denote the number of correctly and incorrectly predicted as changed pixels, and true negative (TN) and false negative (FN) denote the number of correctly and incorrectly predicted as unchanged pixels, respectively. In order to formulate the mentioned five metrics, overall accuracy (OA) and P are calculated by the following equation as a preparation:

$$\begin{cases} \mathrm{OA} = \dfrac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \\ P = \dfrac{(\mathrm{TP}+\mathrm{FP})\cdot(\mathrm{TP}+\mathrm{FN})+(\mathrm{FN}+\mathrm{TN})\cdot(\mathrm{FP}+\mathrm{TN})}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})^2}. \end{cases} \tag{14}$$

Then, these five evaluation metrics could be obtained following the equations in (15). Considering that $F1$, IoU, and Kappa can evaluate the performance of the model in a more comprehensive way, they are given more attention during the experimental analysis

$$\begin{cases} \mathrm{Pre} = \dfrac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \\ \mathrm{Rec} = \dfrac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \\ F1 = \dfrac{2\ \mathrm{Pre}\ \cdot \mathrm{Rec}}{\mathrm{Pre}\ + \mathrm{Rec}} \\ \mathrm{IoU} = \dfrac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}} \\ \mathrm{Kappa} = \dfrac{\mathrm{OA}-P}{1-P}. \end{cases} \tag{15}$$

D. Implementation Details

The proposed CACG-Net is based on PyTorch implementation. Adam [58] is used as an optimizer. A learning rate decay strategy is used. Specifically, as the epoch increases, the learning rate is dynamically adjusted to init_lr$*$ decay_rate$^{(\mathrm{epoch}//\mathrm{decay_epoch})}$. Here init_lr represents the initial learning rate, set to $1e-4$, decay_rate represents the decay rate, set to 0.98, epoch represents the current epoch, and decay_epoch represents the decay period, set to 10. Network training is divided into two phases: training and validation. The training phase is used to update the model parameters and the validation phase is used to save the parameters that currently have the highest IoU on the validation set. Moreover, in order to improve the generalization ability of the model, we apply data enhancement techniques such as random horizontal flipping, random cropping, random rotation, color enhancement, and random noise during the training phase. All experiments are done on an NVIDIA RTX 3090 with the batch size set to 32 for both the training and validation phases.

E. Comparative Experiments

The proposed CACG-Net is compared with eight other CD methods on four public datasets, aiming to validate the feasibility and effectiveness of the proposed method. Next, we will conduct an analysis from two aspects: the quantitative results and the visualization results. For clear observation, TP, TN, FP, and FN are represented by white, black, red, and blue, respectively, in visualization results.

1) Results on LEVIR-CD Dataset: The quantitative results on the LEVIR-CD dataset are displayed in Table I. DSAMNet performs the worst on the LEVIR-CD dataset with the lowest $F1$, IoU, and Kappa of 84.43%, 73.05%, and 83.54%, respectively. It is followed by SRCDNet which has 1.42%, 2.16%, and 1.51% higher $F1$, IoU, and Kappa than DSAMNet, respectively. The performance of FC-Siam-Conc is slightly higher than that of FC-Siam-Diff, which suggests that for the LEVIR-CD dataset, the use of feature concatenation to fuse bitemporal information is more effective. The two methods, BIT and MSCANet, which combine CNN and transformer are superior to the four methods mentioned above. Moreover,

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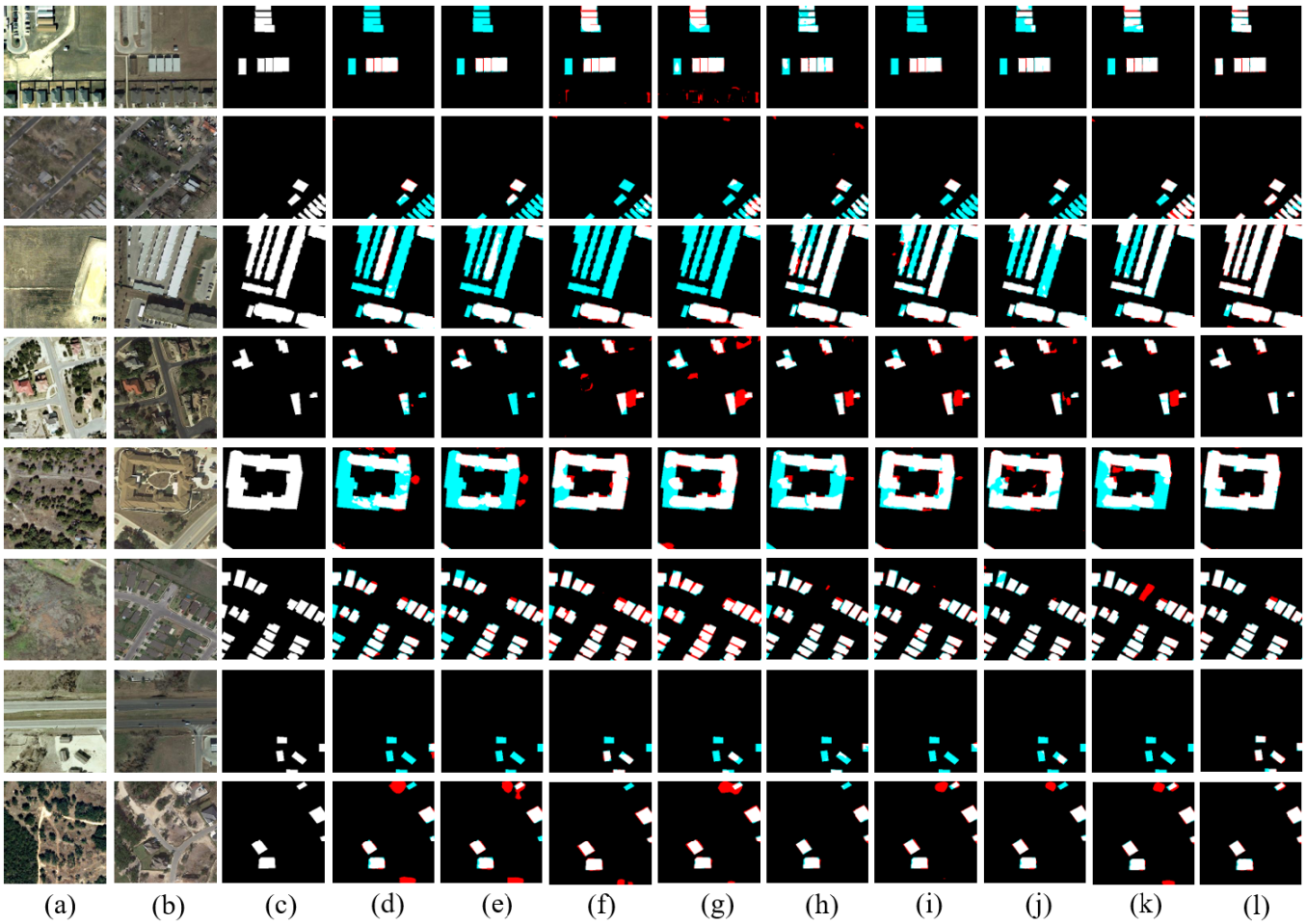


Fig. 8. Visualization results on LEVIR-CD dataset. (a) Image X^1 captured at T_1. (b) Image X^2 captured at T_2. (c) Ground-truth. (d) FC-Siam-Conc. (e) FC-Siam-Diff. (f) SRCDNet. (g) DSAMNet. (h) BIT. (i) MSCANet. (j) DDCNN. (k) DMINet. (l) Ours. White represents TP, black represents TN, red represents FP, and blue represents FN.

we can observe that the three metrics $F1$, IoU, and Kappa are slightly higher for MSCANet than for BIT, which suggests that adding supervision to the deep layer can further improve the model performance. Thanks to the guiding effect of the high-level features, DDCNN has the second-highest performance among the eight comparison methods. DMINet has the highest Pre of 92.64% and it is also second only to the proposed method in terms of $F1$, IoU, and Kappa. This is due to two reasons, one of which is the construction of a joint attention module for enabling the exchange of information between bitemporal features, and the other is the proposal of an incremental aggregation strategy to enable multilevel feature aggregation. Without a doubt, the proposed CACG-Net achieves the best performance with the highest values for Rec, $F1$, IoU, and Kappa, which are 92.41%, 92.29%, 85.68%, and 91.87%, respectively.

Fig. 8 displays the visualization results on the LEVIR-CD dataset, which facilitates us to make a more intuitive comparison. As can be seen from the visualization results, the proposed method is superior to the eight comparison methods, which is consistent with the quantitative comparison results. Whether for small buildings, large buildings, or buildings that are more similar to the background, the proposed method is able to extract change information in a relatively complete manner. This is due to the CAL strategy, which is able to separate the candidate objects from the background, thus maintaining the integrity of the changed objects. In addition, the proposed method performs well in distinguishing between real changes and pseudo-changes. As shown in the third row of Fig. 8, for building color differences caused by changes in lighting, all six comparison methods except FC-Siam-Conc and FC-Siam-Diff identify the building color differences as changes. For small buildings, all eight comparison methods have a miss detection

TABLE I
QUANTITATIVE RESULTS ON LEVIR-CD DATASET

Methods	Pre(%)	Rec(%)	F1(%)	IoU(%)	Kappa(%)
FC-Siam-Conc	89.14	84.66	86.84	76.75	86.16
FC-Siam-Diff	90.05	83.16	86.47	76.16	85.77
SRCDNet	81.49	90.71	85.85	75.21	85.05
DSAMNet	80.23	89.09	84.43	73.05	83.54
BIT	92.16	86.98	89.50	80.99	88.95
MSCANet	90.41	89.10	89.75	81.40	89.20
DDCNN	92.18	89.08	90.60	82.82	90.10
DMINet	**92.64**	89.00	90.78	83.12	90.30
Ours	92.16	**92.41**	**92.29**	**85.68**	**91.87**

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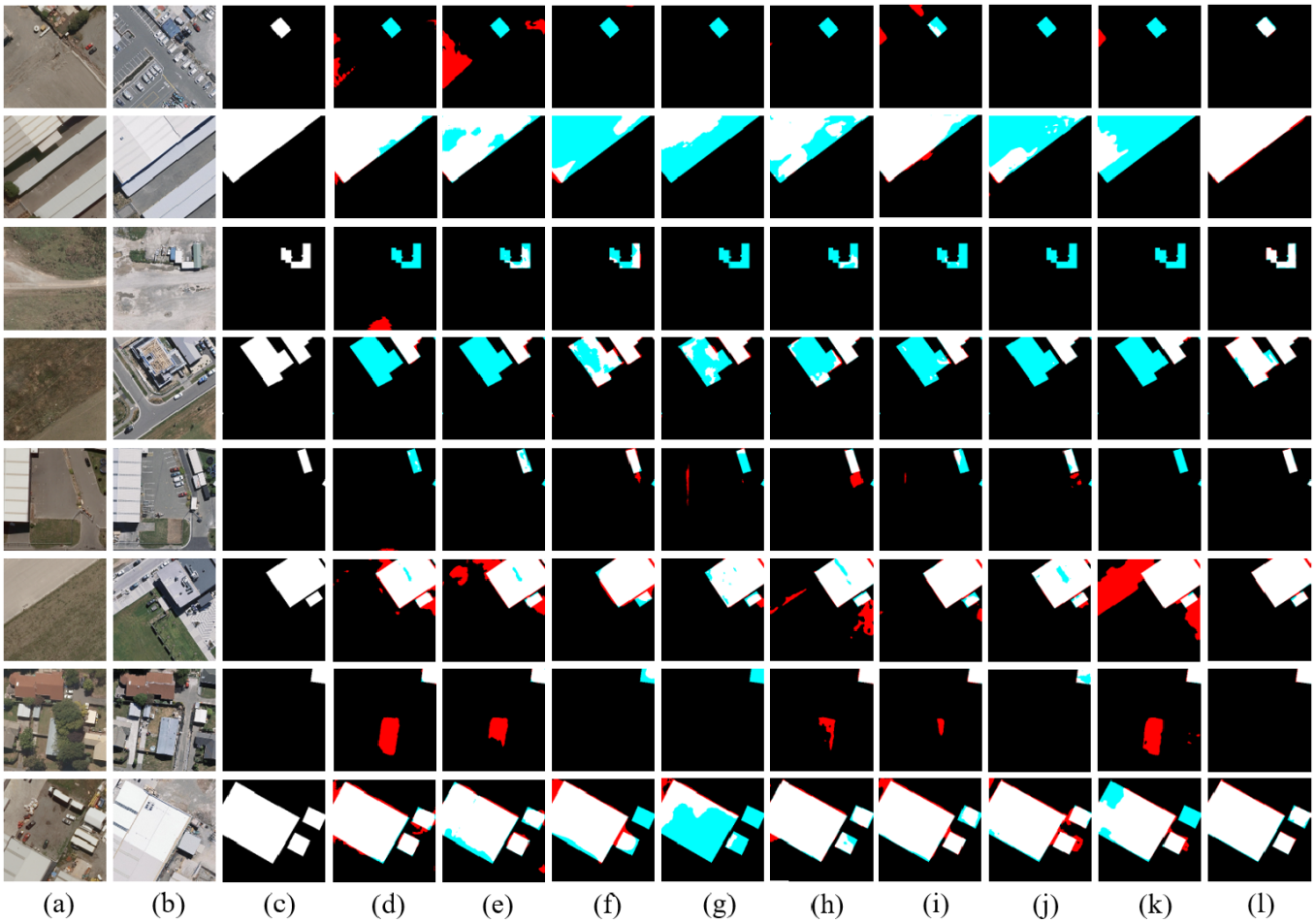


Fig. 9. Visualization results on BCDD dataset. (a) Image X^1 captured at T_1. (b) Image X^2 captured at T_2. (c) Ground-truth. (d) FC-Siam-Conc. (e) FC-Siam-Diff. (f) SRCDNet. (g) DSAMNet. (h) BIT. (i) MSCANet. (j) DDCNN. (k) DMINet. (l) Ours. White represents TP, black represents TN, red represents FP, and blue represents FN.

problem, as shown in the second and seventh rows of Fig. 8, and a false detection problem as well, as shown in the eighth row of Fig. 8. For large buildings, the problem with the eight comparison methods is that the extracted change information is incomplete and the extracted building boundaries are subject to noise interference, as shown in the fifth row of Fig. 8. For building changes that are more similar to the background, the eight comparison methods also perform weakly, as shown in the first row of Fig. 8.

2) Results on BCDD Dataset: Table II displays the quantitative results on the BCDD dataset. As can be seen in Table II, DSAMNet still performs the worst, which indicates its difficulty in recognizing building changes. FC-Siam-Diff has higher $F1$, IoU, and Kappa values than FC-Siam-Conc, which is different from the results on the LEVIR-CD dataset. DMINet has an average performance on the BCDD dataset, on the contrary, SRCDNet gets a good performance. The $F1$, IoU, and Kappa of SRCDNet are 90.69%, 82.96%, and 90.27%, respectively, which is higher than MSCANet by 0.64%, 1.15%, and 0.67%, respectively. The performance of MSCANet on the BCDD dataset is similar to that on the LEVIR-CD dataset, which indicates good generalization performance. DDCNN exhibits performance second only to BIT among the eight comparison methods, with $F1$, IoU, and Kappa values lower than BIT by 1.31%, 2.28%, and 1.36%, respectively. Moreover, DDCNN has the highest Pre of 96.34%. BIT ranks first among the eight comparison methods with $F1$, IoU, and Kappa values of 93.26%, 87.37%, and 92.96%, respectively. The proposed method also achieves the highest performance on the BCDD dataset, obtaining the highest Rec, $F1$, IoU, and Kappa, which are 2.4%, 2.13%, 3.82%, and 2.23% higher than BIT, respectively.

TABLE II
QUANTITATIVE RESULTS ON BCDD DATASET

Methods	Pre(%)	Rec(%)	F1(%)	IoU(%)	Kappa(%)
FC-Siam-Conc	75.40	92.60	83.12	71.12	82.29
FC-Siam-Diff	80.76	91.16	85.65	74.89	84.97
SRCDNet	89.51	91.90	90.69	82.96	90.27
DSAMNet	81.58	78.94	80.23	66.99	79.37
BIT	93.59	92.92	93.26	87.37	92.96
MSCANet	88.20	91.99	90.05	81.81	89.60
DDCNN	**96.34**	87.93	91.95	85.09	91.60
DMINet	87.02	89.19	88.09	78.72	87.55
Ours	95.46	**95.32**	**95.39**	**91.19**	**95.19**

We select eight representative scenarios in the BCDD dataset, including new buildings, building reconstruction, and building disappearance, and their visualization results are displayed in Fig. 9. As can be seen from the figure, the proposed

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TABLE III
QUANTITATIVE RESULTS ON SYSU-CD DATASET

Methods	Pre(%)	Rec(%)	F1(%)	IoU(%)	Kappa(%)
FC-Siam-Conc	80.44	72.92	76.49	61.94	69.70
FC-Siam-Diff	**90.53**	58.18	70.84	54.84	64.24
SRCDNet	73.57	78.60	76.00	61.29	68.27
DSAMNet	74.88	82.16	78.35	64.41	71.26
BIT	81.90	76.19	78.94	65.21	72.74
MSCANet	81.33	75.70	78.42	64.49	72.06
DDCNN	81.82	78.88	80.32	67.12	74.40
DMINet	84.13	79.76	81.89	69.33	76.49
Ours	82.42	**84.30**	**83.35**	**71.45**	**78.13**

method has fewer false detections and missed detections, extracts more complete information about building changes, and does a better job of retaining information about edge details. For new buildings, a common problem with the eight comparison methods is missed detection, as shown in the first, third, and fourth rows of Fig. 9. For complex and large new buildings, they often suffer from false detection, incorrectly identifying roads as building changes. Among them, DMINet has the most serious problem of false detection, as shown in the sixth row of Fig. 9. For large building reconstruction, the change information they extracted is incomplete, as shown in the second row of Fig. 9. For building disappearance, SRCDNet, DSAMNet, and DMINet do not detect it at all, as shown in the eighth row of Fig. 9. Moreover, for buildings that are shaded by trees, all five comparison methods recognize them as changes, except for SRCDNet, DSAMNet, and DDCNN, as shown in the seventh row of Fig. 9.

3) Results on SYSU-CD Dataset: Similar to the results on the above two datasets, the proposed method also achieves optimal results on the SYSU-CD dataset, the results of which are presented in Table III. The proposed method achieves the highest values for the four metrics Rec, $F1$, IoU, and Kappa, which are 84.30%, 83.35%, 71.45%, and 78.13%, respectively. DMINet ranks second, with its $F1$, IoU, and Kappa metrics second only to the proposed method. It is worth noting that on the SYSU-CD dataset, DDCNN and BIT also achieve better performance, which confirms their good generalization performance. MSCANet and DSAMNet have similar performance on the SYSU-CD dataset. FC-Siam-Conc, SRCDNet, and FC-Siam-Diff perform poorly, and FC-Siam-Diff has the worst performance. FC-Siam-Diff has the lowest $F1$, IoU, and Kappa values of 70.84%, 54.84%, and 64.24%, respectively, which suggests that the bitemporal information fusion approach using feature differences is prone to the loss of effective information when confronted with complex scenarios with multiple categories of changing information.

In contrast to the two datasets mentioned above, the SYSU-CD dataset contains information on changes in multiple categories, which increases the complexity of the scenarios. Fig. 10 displays the visualization results on the SYSU-CD dataset. It can be seen that facing the complex scenario of the SYSU-CD dataset, the proposed method also achieves better visualization results than the eight comparison methods, with fewer false and missed detection problems. The visualization results of the eight comparison methods are poor, and there are multiple problems when facing changes in complex scenes. Except for FC-Siam-Diff, the remaining seven comparison methods face varying degrees of false detection problems as shown in the second, third, and fourth rows of Fig. 10. The main problem of FC-Siam-Diff is missed detection, which corresponds to the highest Pre and the lowest Rec in Table III, as shown in Fig. 10(e). When confronted with large-scale changes due to construction, the information on changes extracted by the eight comparison methods is incomplete, as shown in the first and sixth rows of Fig. 10. In addition, as shown in the fifth row of Fig. 10, the edges produced by the eight comparison methods are relatively rough, whereas the proposed method can produce sharp edges. This is due to the proposed strategy of gradually revealing changed objects in a coarser-to-finer way, preserving important changed objects on a coarse scale, and revealing change details on a fine scale.

4) Results on CLCD Dataset: On the CLCD dataset, the proposed method obtains the highest values on all five evaluation metrics, where the $F1$, IoU, and Kappa values are higher than the second-ranked MSCANet by 6.93%, 8.9%, and 7.39%, respectively, and the results are displayed in Table IV. BIT has the second-highest performance among the eight comparison methods, with DMINet following closely behind. The good performance of MSCANet and BIT again demonstrates the effectiveness of combining CNN and transformer. FC-Siam-Conc and FC-Siam-Diff have similar performances. Unlike the results on the previous datasets, DDCNN performs poorly on the CLCD dataset with $F1$, IoU, and Kappa values of 56.81%, 39.67%, and 53.66%, respectively. SRCDNet and DSAMNet have the worst performance, suggesting that metric-based learning strategies are weak in the face of multicategory complex scenarios with the CLCD dataset.

Similar to the SYSU-CD dataset, the CLCD dataset also has complex scenarios, where we chose several representative scenarios of roads, lakes, buildings, and bare soil lands, and their visualization results are displayed in Fig. 11. From the visualization results, it can be seen that the visualization of the eight comparison methods is weaker than our method, and they have more missed and false detection problems, which corresponds to the quantitative results shown in Table IV. The strength of our method is reflected in the continuity of the extracted roads and the integrity of the extracted buildings, lakes, and bare soil lands. For the identification of road changes, our method outperforms the comparison methods in terms of maintaining road continuity, even though our method still falls short in road boundary details. SRCDNet and DSAMNet have more serious false detection problems, while DDCNN does not identify the road changes at all, as shown in the sixth row of Fig. 11. For the identification of building changes, false detection is a big problem faced by the eight comparison methods, as shown in the third and eighth rows of Fig. 11. Similarly, for the identification of bare soil land changes, the eight comparison methods suffer from the problem of false detection. In addition to that, the change information they extracted is very incomplete, especially for FC-Siam-Conc, FC-Siam-Diff, and DDCNN, where they did not detect the change at all, as shown in the second row of Fig. 11. For the identification of lake changes, only our

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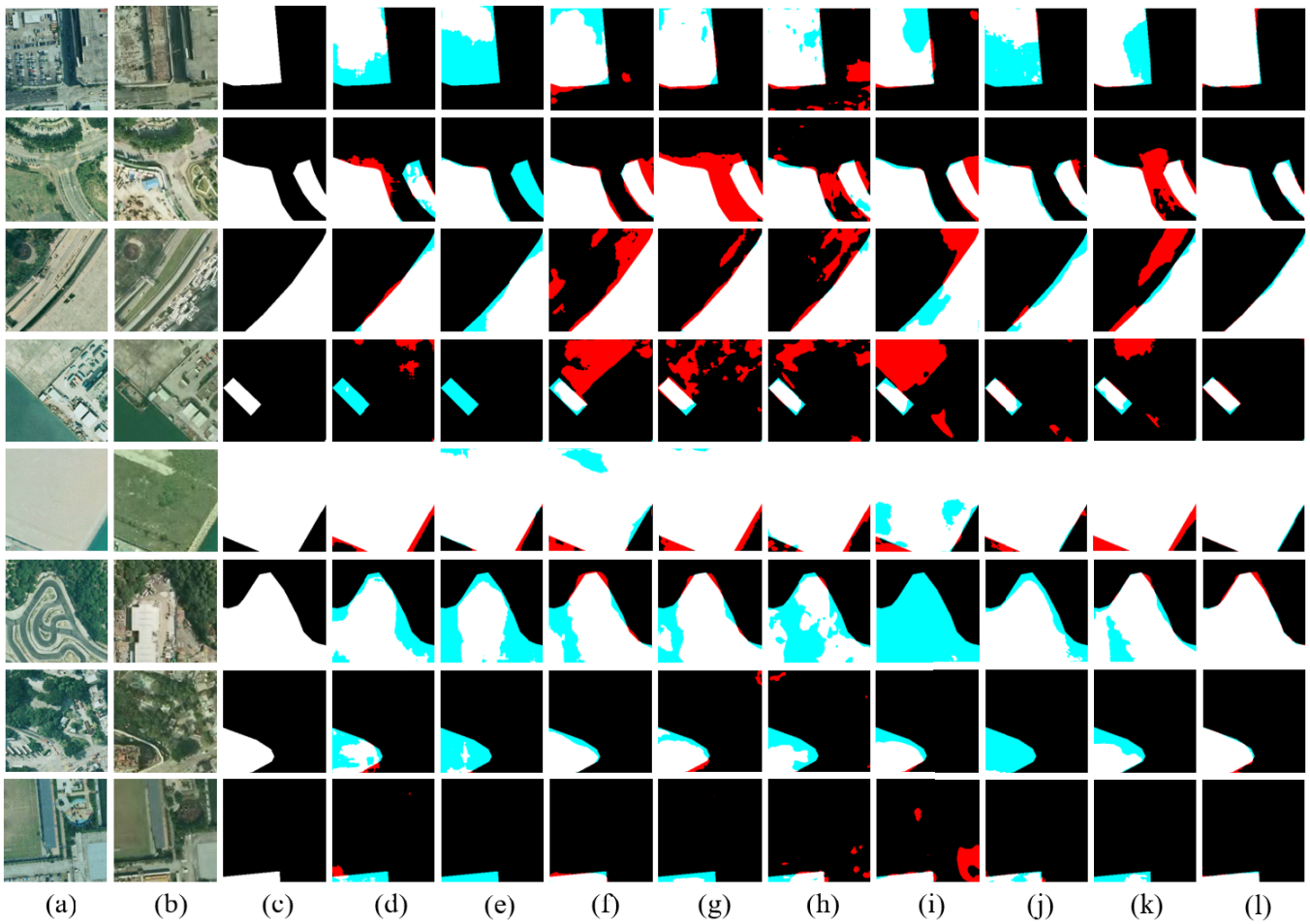


Fig. 10. Visualization results on SYSU-CD dataset. (a) Image X^1 captured at T_1. (b) Image X^2 captured at T_2. (c) Ground-truth. (d) FC-Siam-Conc. (e) FC-Siam-Diff. (f) SRCDNet. (g) DSAMNet. (h) BIT. (i) MSCANet. (j) DDCNN. (k) DMINet. (l) Ours. White represents TP, black represents TN, red represents FP, and blue represents FN.

method is able to obtain relatively complete information about their changes, while the performance of the eight comparison methods is poor, as shown in the seventh row of Fig. 11.

F. Ablation Study

An ablation study of the proposed CACG-Net is performed on four public datasets to verify the effectiveness of the three key components, MCD, CAL, and CGM, as well as increasing the edge weights (denoted by W_e) in the training procedure. As a special note, the output Sub of CAL is not utilized to calibrate the change results produced by CGM when the role of CGM is verified alone. It is when combining CGM and CAL that the output Sub of CAL is utilized. We remove MCD, CAL, and CGM from CACG-Net to be used as the "Base" model. Specifically, feature concatenation is used to fuse the bitemporal information obtained from the ResNet-18 network, and then a decoder is constructed to hierarchically aggregate the multilevel features, where the decoder consists of convolutional layers with 3×3 and 1×1 kernels. Based on the "Base" model, multiple sets of experiments are set up to verify the effectiveness of each component and the combination of them, and the results of the ablation experiments on the four public datasets are presented in Table V. The detailed analyses are as follows.

TABLE IV
QUANTITATIVE RESULTS ON CLCD DATASET

Methods	Pre(%)	Rec(%)	F1(%)	IoU(%)	Kappa(%)
FC-Siam-Conc	65.61	59.27	62.28	45.22	59.41
FC-Siam-Diff	71.90	54.62	62.08	45.01	59.48
SRCDNet	40.48	61.15	48.71	32.20	43.67
DSAMNet	46.35	64.17	53.82	36.82	49.46
BIT	72.71	65.76	69.06	52.75	66.71
MSCANet	75.74	68.04	71.68	55.86	69.54
DDCNN	62.30	52.20	56.81	39.67	53.66
DMINet	70.56	62.12	66.07	49.33	63.53
Ours	**80.61**	**76.71**	**78.61**	**64.76**	**76.93**

1) Effectiveness of MCD: MCD is used to reveal the most important changed objects in the coarse shape from the basic features extracted by ResNet-18. Comparing the first and second rows of Table V, it can be seen that the introduction of MCD into the "Base" model achieves significant performance improvement on the four public datasets. On the LEVIR-CD dataset, the introduction of MCD improves the $F1$, IoU, and Kappa of the "Base" model by 0.52%, 0.9%, and 0.56%, respectively. The same effect is also observed on the remaining three datasets, and the specific results are shown in Table V. The above results demonstrate the effectiveness of MCD in CACG-Net.

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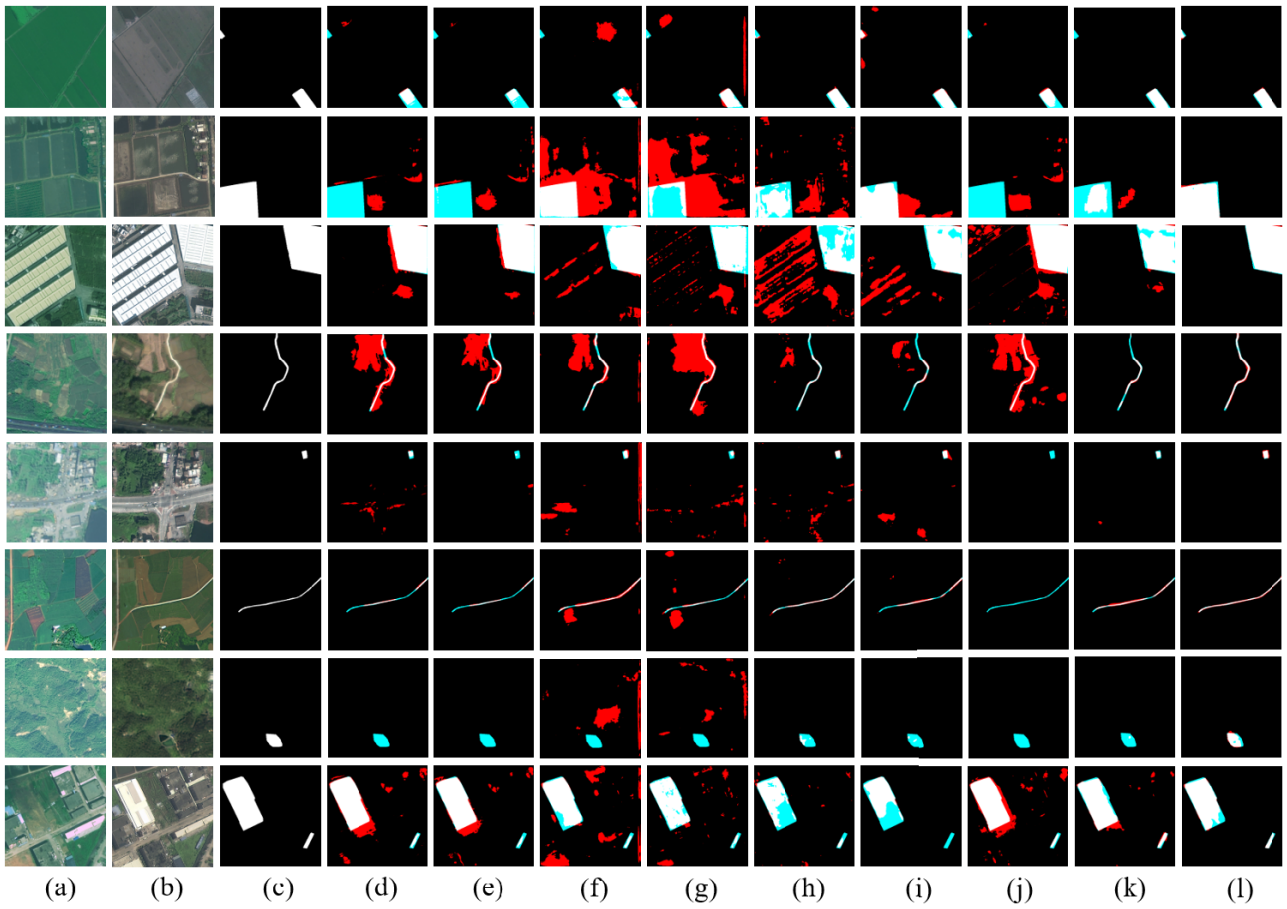


Fig. 11. Visualization results on CLCD dataset. (a) Image X^1 captured at T_1. (b) Image X^2 captured at T_2. (c) Ground-truth. (d) FC-Siam-Conc. (e) FC-Siam-Diff. (f) SRCDNet. (g) DSAMNet. (h) BIT. (i) MSCANet. (j) DDCNN. (k) DMINet. (l) Ours. White represents TP, black represents TN, red represents FP, and blue represents FN.

TABLE V

ABLATION STUDY ON FOUR PUBLIC DATASETS. ALL RESULTS ARE EXPRESSED AS A PERCENTAGE (%)

	MCD	CGM	CAL	W_e	LEVIR-CD			BCDD			SYSU-CD			CLCD		
					F1	IoU	Kappa	F1	IoU	Kappa	F1	IoU	Kappa	F1	IoU	Kappa
Base					91.24	83.88	90.76	93.94	88.58	93.67	80.59	67.49	74.66	70.98	55.01	68.49
Base	√				91.76	84.78	91.32	94.67	89.88	94.43	81.07	68.17	74.93	73.62	58.25	71.61
Base		√			91.87	84.97	91.44	94.55	89.66	94.31	80.99	68.05	74.90	74.41	59.25	72.48
Base			√		91.85	84.92	91.40	94.68	89.89	94.44	81.52	68.81	75.72	74.44	59.29	72.18
Base				√	91.72	84.71	91.28	94.63	89.81	94.40	80.79	67.77	75.06	72.74	57.16	70.58
Base	√	√			91.97	85.13	91.54	94.90	90.30	94.68	81.62	68.95	76.02	76.34	61.73	74.42
Base	√	√	√		92.16	85.46	91.74	95.02	90.51	94.80	83.17	71.18	78.01	77.60	63.40	75.74
Base	√	√	√	√	92.29	85.68	91.87	95.39	91.19	95.19	83.35	71.45	78.13	78.61	64.76	76.93

2) Effectiveness of CGM: CGM adopts a coarser-to-finer way to gradually reveal the changed objects in response to the fact that the changed objects exhibit coarse/fine shapes in different scales. The third row of Table V shows the results of introducing CGM in the "Base" model. On the LEVIR-CD dataset, the $F1$ of the "Base" model is improved from 91.24% to 91.87%, the IoU from 83.88% to 84.97%, and the Kappa from 90.76% to 91.44%. On the BCDD dataset, the $F1$, IoU, and Kappa of the "Base" model improve by 0.61%, 1.08%, and 0.64%, respectively. In addition, on the SYSU-CD and CLCD datasets, the introduction of CGM also significantly improves the performance of the "Base" model. The above results show that it is effective to adopt a coarser-to-finer way of revealing changed objects.

3) Effectiveness of CAL: CAL is designed to select candidate objects in each temporal image in response to the fact that a certain change usually corresponds to a real object in either of the two images. Comparing the first and fourth rows of Table V, it is easy to find that the introduction of CAL makes the "Base" model improve the $F1$, IoU, and Kappa values on all four datasets, with the biggest improvement on the CLCD dataset. The $F1$, IoU, and Kappa of the "Base" model are improved by 3.46%, 4.28%, and 3.69%, respectively, on the CLCD dataset. These experimental results show that selecting candidate objects on each temporal image is effective for correctly recognizing changes.

4) Effectiveness of W_e: The increasing edge weights strategy is used to increase the attention to the edge information of the changed areas in the training procedure, which is effective

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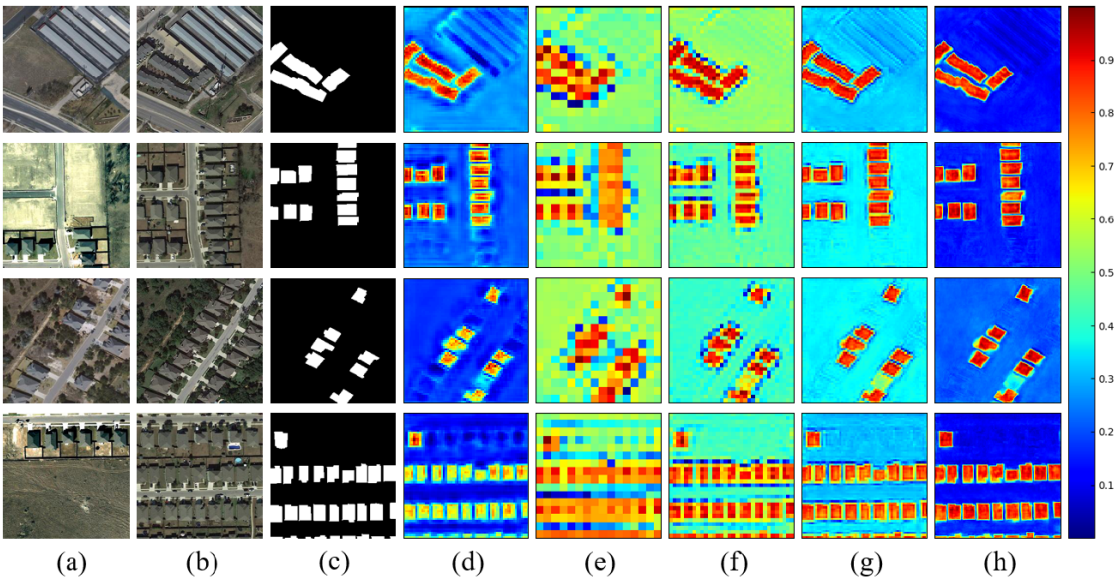


Fig. 12. Visualization results of MCD and CGM processes on the LEVIR-CD dataset. (a) Image X^1 captured at T_1. (b) Image X^2 captured at T_2. (c) Ground-truth. (d) DI_6 generated by MCD. (e) DI_5 generated by CGM in Stage5. (f) DI_4 generated by CGM in Stage4. (g) DI_3 generated by CGM in Stage3. (h) DI_2 generated by CGM in Stage2.

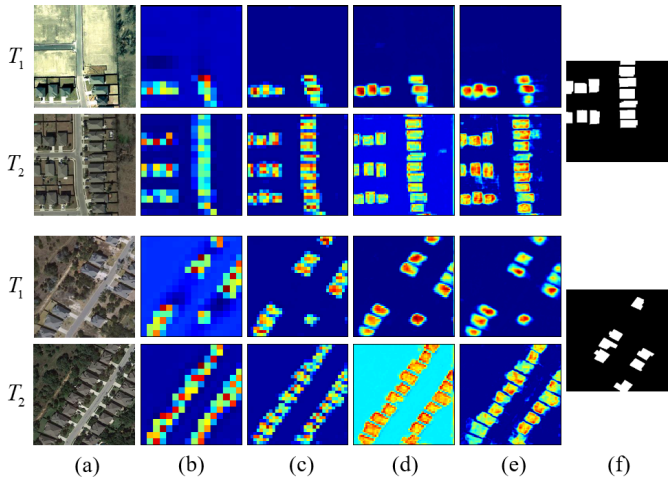


Fig. 13. Visualization results of the proposed CAL process on the LEVIR-CD dataset. Note that the first two (last two) rows indicate the first (second) pair of bitemporal images and their segmentation results. (a) Original bitemporal images. (b) Segmentation results generated by CAL in Stage5. (c) Segmentation results generated by CAL in Stage4. (d) Segmentation results generated by CAL in Stage3. (e) Segmentation results generated by CAL in Stage2. (f) Corresponding CD Ground-truth.

for correctly detecting the changes, and its usefulness is verified by the experimental results in Table V. The application of W_e results in $F1$, IoU, and Kappa values of 91.72%, 84.71%, and 91.28%, respectively, on the LEVIR-CD dataset. On the BCDD dataset, it achieves $F1$, IoU, and Kappa values of 94.63%, 89.81%, and 94.40%, respectively. On the SYSU-CD dataset, it achieves $F1$, IoU, and Kappa values of 80.79%, 67.77%, and 75.06%, respectively. As for the CLCD dataset, it achieves $F1$, IoU, and Kappa values of 72.74%, 57.16%, and 70.58%, respectively.

5) Effectiveness of Combinations Between Them: The last three rows of Table V show the results of experiments in which MCD, CGM, CAL, and W_e are sequentially combined. It can be seen that "Base+MCD+CGM" has higher $F1$, IoU, and Kappa than "Base+MCD" on the four datasets. The $F1$, IoU, and Kappa values of "Base+MCD+CGM+CAL" are also higher than those of "Base+MCD+CGM." Introducing MCD, CGM, CAL, and W_e into the "Base" model at the same time achieves the optimal results on all four datasets, which shows that the combination of them is able to realize the gain effect.

G. Visualization of Learning Process

The learning process of the proposed MCD, CGM, and CAL are visualized to provide a more intuitive exemplification, where Figs. 12 and 13 show the results of the corresponding visualization on the LEVIR-CD dataset. A detailed description follows.

1) Visualization of MCD and CGM: MCD is used to reveal the most important changed objects in the coarse shape, which is verified in column Fig. 12(d). We can see that some important features are captured after MCD. CGM adopts a coarser-to-finer way to gradually reveal the changed objects, and Fig. 12(e)–(h) illustrates this process. From Fig. 12(e)–(h), we can observe that the features are coarse-grained to fine-grained, and the change details are gradually enriched. In addition, Fig. 12(h) has the richest details of the changes, which validates the effectiveness of adopting a coarser-to-finer learning way.

2) Visualization of CAL: CAL is used to select candidate objects in each temporal image, and Fig. 13 shows the candidate objects selected on bitemporal images. It can be observed from the figure that the candidate objects selected by the CAL include two categories, changed objects as well as unchanged objects and the final changed objects are included in the selected candidate objects, which shows that the CAL is helpful in searching for changes of interest.

H. Learning Curve Comparison

We visualize the learning curves of the proposed CACG-Net with four comparison methods, DMINet, DDCNN, MSCANet,

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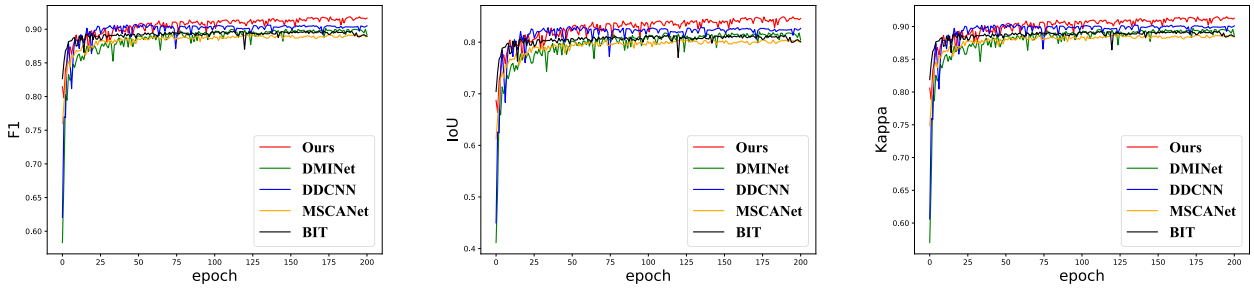


Fig. 14. Learning curve visualization. (a) $F1$ value of LEVIR-CD validation set. (b) IoU value of LEVIR-CD validation set. (c) Kappa value of LEVIR-CD validation set.

TABLE VI
ANALYSIS RESULTS OF PARAMETER β_l

β_5	β_4	β_3	β_2	F1(%)	IoU(%)	Kappa(%)
0.1	0.2	0.3	0.4	91.61	84.53	91.16
0.25	0.25	0.25	0.25	91.98	85.16	91.56
0.4	0.3	0.2	0.1	92.05	85.27	91.62
0.5	0.3	0.2	0	92.09	85.35	91.67
0.5	0.4	0.1	0	92.13	85.42	91.71
0.5	0.4	0	0.1	92.16	85.45	91.73
0.5	0.5	0	0	92.18	85.50	91.76
0.6	0.3	0.1	0	92.07	85.30	91.64
0.6	0.3	0	0.1	92.16	85.46	91.73
0.6	0.4	0	0	92.22	85.56	91.80
0.7	0.3	0	0	**92.29**	**85.68**	**91.87**
0.8	0.2	0	0	92.11	85.37	91.68

and BIT. Fig. 14 shows their $F1$, IoU, and Kappa values for the first 200 epochs on the LEVIR-CD validation set. By comparing the learning curves, it can be observed that the proposed CACG-Net obtains good performance in terms of model stability, convergence speed, and detection accuracy. MSCANet has the optimal model stability, but its $F1$, IoU, and Kappa values are low. DDCNN has a faster convergence speed, but its model stability is weak. DMINet also has weak model stability. BIT is better than DDCNN and DMINet in terms of model stability, but has the problem of having lower $F1$, IoU, and Kappa values.

V. DISCUSSION

In this section, we discuss the weights β_l of CAL loss at different stages and analyze the model efficiency of the proposed CACG-Net and other comparison methods. In addition, we explore the performance of different backbones and conduct a comparative analysis of loss functions.

A. Parametric Analysis

Table VI presents the results of the analysis for the parameter β_l. Here β_5 (i.e., $l = 5$) represents the weight of CAL loss at the fifth stage, likewise β_4, β_3, and β_2. From the table, it can be observed that the higher the values of β_5 and β_4, the better the performance of the model, which indicates that the higher the weight of the CAL loss in the higher stages, the better it is for the model. The model achieves optimal performance when β_5 is equal to 0.7, β_4 is equal to 0.3, and the CAL loss weights are 0 for the other two stages. Therefore, we only perform the CAL process at the highest two stages.

B. Model Efficiency

In this article, the model efficiency of different methods is evaluated from the perspectives of parameters quantity (Params) and floating-point operations (FLOPs), aiming to understand their practical application requirements. Params and FLOPs are used to measure the size and computational complexity of a model, respectively. Params is measured in units of 10^6 (M), and FLOPs is measured in units of 10^9 (G).

Fig. 15 displays the Params and FLOPs of the proposed CACG-Net and eight comparison methods. Note that the image size used in the experiment is $1 \times 3 \times 256 \times 256$. From Fig. 15, it is intuitively found that FC-Siam-Conc and FC-Siam-Diff have the lowest Params and FLOPs, while DDCNN has the highest values in both metrics. SRCDNet and DSAMNet also have higher Params and FLOPs, and it is worth noting that their Kappa values on the LEVIR-CD dataset are lower than those of the other methods. DMINet achieves higher performance on the LEVIR-CD dataset with lower Params and FLOPs. The proposed CACG-Net also has the same effect as DMINet, achieving the highest Kappa with lower Params and FLOPs. In addition, we also observe that the proposed CACG-Net has similar Params and FLOPs with BIT, but the Kappa achieved by CACG-Net is much higher than that of BIT. The above results indicate that the proposed CACG-Net exhibits higher model efficiency.

C. Comparison of Backbones

The proposed CACG-Net utilizes ResNet-18 as a backbone to acquire multilevel features in bitemporal images. However, to further validate the generalizability of the proposed MCD, CAL, CGM, and increasing edge weights strategy, we adopt different backbones for comparative analysis. Specifically, we select several backbones commonly used in the field of CD, i.e., MobileNet-V2 [59], VGG-16 [60], and Res2Net [61], for comparison with ResNet-18. The results are displayed in Table VII.

As can be seen from Table VII, MobileNet-V2, VGG-16, and Res2Net all achieve good performance. They all outperform the eight comparison methods on the LEVIR-CD dataset, which validates the generalizability of the proposed modules. MobileNet-V2 is a lightweight backbone that has lower performance on the LEVIR-CD dataset than the other three backbones, but it has the lowest Params and FLOPs at 3.85M and 11.47G, respectively. The $F1$, IoU, and Kappa values of VGG-16 are 92.16%, 85.46%, and 91.74%, respectively,

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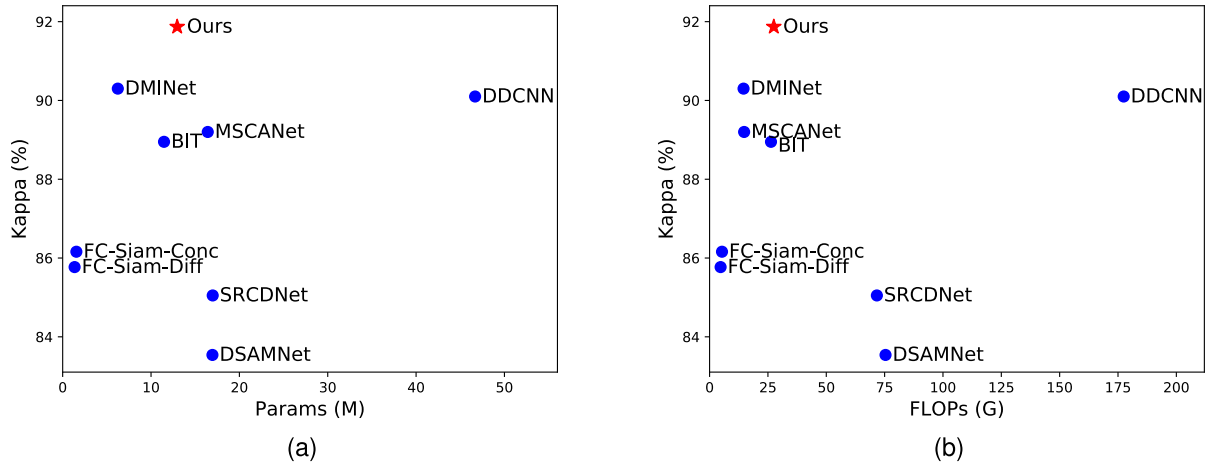


Fig. 15. Comparison of model efficiency. (a) Params of different methods. (b) FLOPs of different methods.

TABLE VII
COMPARISON BETWEEN DIFFERENT BACKBONES

Backbones	Params (M)	FLOPs (G)	LEVIR-CD F1(%)	LEVIR-CD IoU(%)	LEVIR-CD Kappa(%)
MobileNet-V2	3.85	11.47	92.01	85.21	91.58
VGG-16	16.65	49.28	92.16	85.46	91.74
Res2Net	26.37	57.86	92.36	85.80	91.95
ResNet-18(Ours)	12.96	27.51	92.29	85.68	91.87

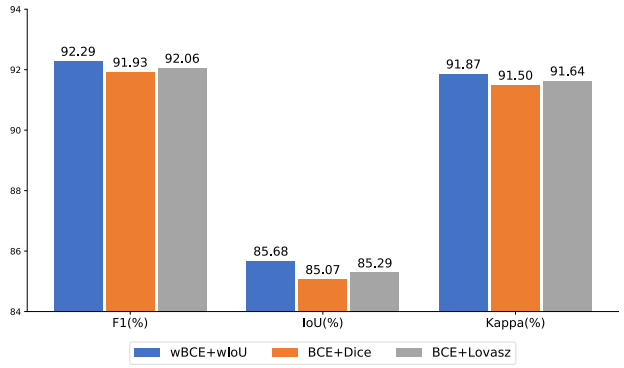


Fig. 16. Comparison between different loss functions.

which are better than that of MobileNet-V2, but its Params and FLOPs are much higher than that of MobileNet-V2. Res2Net achieves the highest performance, but it also has the highest Params and FLOPs, 26.37M and 57.86G, respectively, which makes it difficult to strike a balance between accuracy and efficiency. ResNet-18 has the second highest performance, slightly lower than Res2Net, and also has lower Params and FLOPs than Res2Net and VGG-16, enabling a balance between accuracy and efficiency. Therefore, we adopt ResNet-18 as the backbone of CACG-Net in our experiments.

D. Comparison of Loss Functions

Given that a single loss may have limitations in a specific scenario, a hybrid loss strategy can effectively utilize the complementary advantages of different loss functions to compensate for the shortcomings of a single loss. Therefore, in this article, we adopt a combination of wBCE and wIoU loss. Previous works have similar schemes, such as BCE+Dice loss [62] and BCE+Lovasz loss [63]. Next, we compare the wBCE+wIoU loss with these two types of hybrid losses. The BCE+Dice loss effectively mitigates the problem of class imbalance, but it has the disadvantages of high optimization difficulty and weaker recognition of small targets. The BCE+Lovasz loss is sensitive to boundaries, facilitating the generation of precise boundaries, but its drawbacks are high computational cost and low training stability. Compared to the aforementioned two hybrid losses, the wBCE+wIoU loss has advantages in three aspects. First, it directly correlates with the IoU evaluation metric, benefiting the improvement of model performance. Second, it adds weight to the edges of changed areas, facilitating the generation of precise boundaries, and also alleviates the class imbalance between changed and unchanged pixels. Third, it exhibits better generalizability and stability.

Fig. 16 shows the results of the three hybrid losses on the LEVIR-CD dataset. As can be seen from the figure, the wBCE+wIoU loss achieves the best performance, with its $F1$, IoU, and Kappa values being 0.23%, 0.39%, and 0.23% higher, respectively, than those of the second-ranked BCE+Lovasz loss. The BCE+Dice has the poorest performance, with its $F1$, IoU, and Kappa values being 91.93%, 85.07%, and 91.50%, respectively.

VI. CONCLUSION

In this article, a coarser-to-finer CD method, called CACG-Net, is proposed for the changed objects exhibiting coarse/fine shapes in different scales. It contains three key components, namely, MCD, CAL, and CGM. MCD is designed to combine information from the highest three stages to reveal the most important changed objects in the coarse shape with less computational cost. CAL is used to select candidate objects on each temporal image and associate them with the change of interest through variant change losses. The way of revealing changed objects from coarser to finer is realized by CGM, which progressively enriches the change details by combining coarser change results with finer features. Moreover, in the training procedure, the edges of changed objects are emphasized by increasing the corresponding weights in wBCE and wIoU to get clear boundaries. A series of experiments on four popular datasets demonstrate the effectiveness

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of CACG-Net. Compared to eight state-of-the-art CD methods, CACG-Net achieves optimization in terms of accuracy and efficiency. However, it is noticed that supervised CD task relies on huge pixel-level labeled samples, which requires plenty of manual labor and highly limits the practical application of CD techniques. Therefore, the development of semi-supervised, weakly-supervised, or unsupervised CD techniques is very necessary, and we would like to make the best effort on these tasks in future work.

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